

Which Kind of Law? Applying a Nomological Distinction to a Claimed Retrocausal X-ray Experiment

A synthesis applying the a priori / a posteriori nomology distinction to *Demonstrating Retrocausal Wavefunction Collapse and Latent Momentum Uncertainty via Spatial Post-Selection of Single X-ray Photons* (Wagenknecht, 2026)

1. Purpose of This Synthesis

The companion paper 'On the Non-Identity of A Priori and A Posteriori Nomologies of Physical Motion' argues that two systems of physical law can describe the same motions and still be structurally non-alike, because they differ in modal status, justificatory structure, and response to anomaly. This note applies that three-part diagnostic to the X-ray retrocausality write-up, treating the latter as a candidate 'nomology' of photon behavior and asking which kind of system it is actually presenting itself as — and whether it is internally consistent in that presentation.

2. What the Experiment Paper Claims to Show

The write-up describes a single-photon X-ray interferometer (beryllium double slit, asymmetric Bragg crystal, 50/50 beam splitter, a 1.5-micron 'spatial shift aperture' on the reflected arm, and a NaI scintillation detector scanning the far field). Its claimed result is that late, restrictive position measurement at the aperture 'retroactively dictates' which slit the photon went through, and that this constraint also produces a 'latent' momentum spread that later shows up as a diffraction pattern. The paper frames this in the language of the Two-State Vector Formalism (TSVF), under which a quantum system's description is fixed jointly by an initial state and a final (post-selected) state.

3. Applying Axis (a): Modal Status

The companion paper distinguishes laws treated as necessary relative to a framework from laws treated as contingent, revisable generalizations. The X-ray paper equivocates between these registers. It presents its central claim — that post-selection 'retroactively dictates' the earlier path — in the constitutive language of an a priori system: the collapse is described as following from the formal structure of TSVF itself, prior to and independent of what any specific run of the apparatus would show. Yet the paper also offers this as an empirical, falsifiable result to be confirmed by 'Monte Carlo simulations' and detector counts, which is the posture of an a posteriori system. On the framework developed in the companion paper, these are not two descriptions of one claim; a claim cannot simultaneously be true in virtue of formal necessity and true in virtue of contingent, observation-answerable regularity. The paper does not resolve which kind of claim it is making, and so does not establish the modal status of its own central result.

4. Applying Axis (b): Justificatory Structure

In an a posteriori nomology, 'what observation would refute this?' must have a non-trivial answer. For the X-ray paper, this question is hard to answer in the form given. TSVF, as standard quantum mechanics, is empirically equivalent to ordinary quantum mechanics for any completed experiment; the formalism is a redescription of standard unitary evolution plus projective measurement, not a separate causal channel running backward through time. Consequently, no detector statistics — including the far-field click pattern the paper treats as confirmation — distinguish 'retrocausal collapse' from the ordinary explanation that post-selecting on the reflected, narrowly-apertured branch simply conditions

which subset of photons get counted, reshaping the recorded distribution without any backward action on the photon's earlier history. The paper does not identify what a disconfirming result would look like, which is the diagnostic the companion paper flags as marking an a priori-style claim wearing a posteriori clothing.

5. Applying Axis (c): Response to Anomaly

The companion paper notes that a priori systems tend to absorb counter-instances by reinterpreting the case rather than revising the law. The X-ray paper's own diagram labels the beam splitter a 'chance box' with a stipulated '50/50 chance of retrocausality' — that is, the retrocausal interpretation is built into the apparatus description by definition, rather than being a hypothesis the apparatus could fail to support. If, on a real run, the far-field pattern matched ordinary post-selected diffraction (the standard-physics expectation), the paper's framing leaves no room to record that as evidence against retrocausal collapse, since the label 'retrocausal' has already been assigned to the mechanism by stipulation rather than earned through a result that could have come out otherwise.

6. What Kind of Document This Is

Read through the companion paper's lens, the X-ray write-up is best understood not as an a posteriori nomology — a generalization open to refutation by the data it describes — but as a stipulative narrative dressed in the instrumentation and statistical vocabulary of one. It assigns a contested theoretical interpretation (genuine backward-in-time influence) to a setup (post-selected interferometry) whose textbook account does not require or support that interpretation, and it does so without specifying an observational outcome that would have counted against it. The companion paper's qualification in Section 5 — that the a priori/a posteriori line is not always fixed in advance — does not rescue this case, since the issue here is not a legitimate stage-dependent shift in a principle's status, but an unmarked slide between the two postures within a single argument.

7. Summary

Treating 'laws of motion' (or, by extension, laws governing photon trajectories) as interchangeable regardless of their a priori or a posteriori standing obscures exactly the kind of conflation seen here: a result presented with the rhetorical confidence of formal necessity, but the experimental trappings of contingent, testable science, with neither register's standards — internal coherence for the former, falsifiability for the latter — fully met.